

Quick Reference Guide

Module 8: Capstone: Building and Defending a Production-Grade RAG System

Learning Outcomes:

- Present the architecture, behavior, and evaluation results of a production-grade retrieval-augmented LLM system.
- Justify engineering design decisions by explaining the trade-offs among retrieval strategies, evaluation approaches, security, performance, and operational considerations.

Key Concepts

Preparing for a Design Review

An effective design review focuses on telling the engineering story behind your system rather than demonstrating technical perfection.

The purpose of the review is to:

- Explain how your system works
- Justify your design choices with evidence
- Communicate key trade-offs and limitations
- Identify opportunities to strengthen your final presentation

Telling the Engineering Story of Your System

Organize your presentation around five key questions:

1. What problem does your system solve?

Explain:

- The problem or opportunity
- The intended users

- The types of questions or tasks your system supports

2. How does your system work?

Describe:

- Overall architecture
- Major workflow components
- Retrieval approach
- How retrieval, reasoning, and generation interact
- How the system behaves in representative retrieval and generation scenarios

Focus on the overall design rather than implementation details.

3. How do you know your system works?

Support your conclusions with evidence.

Examples include:

- Evaluation methodology
- Representative outputs
- Successful use cases
- Known limitations
- Testing results

Evidence should support the conclusions you present.

4. Why did you make those design choices?

Explain:

- Major architectural decisions
- Retrieval strategy

- Alternatives considered
- Why your final approach was selected

Explain the reasoning behind your design choices and support them with evidence gathered throughout development.

5. What would you improve next?

No system is perfect.

Reflect on:

- Remaining limitations
- Future improvements
- Lessons learned
- Additional evaluation that could strengthen the system

Explaining Engineering Trade-Offs

Be prepared to discuss trade-offs such as:

- Performance vs. cost
- Security vs. flexibility
- Latency vs. output quality
- Simplicity vs. capability
- Retrieval accuracy vs. response speed

Strong presentations explain why a particular balance was appropriate for the intended use case.

Giving Effective Peer Feedback

Constructive feedback should be:

- ✓ Specific
- ✓ Actionable
- ✓ Evidence-based

Focus on:

- Clarity of explanation
- Strength of supporting evidence
- Engineering reasoning
- Opportunities for improvement
- Communication of key trade-offs

Avoid comments that provide little guidance (for example, “Looks good” or “Nice job”).

Receiving Feedback

When participating in a design review:

- Listen before responding
- Ask clarifying questions
- Consider how your explanation could be strengthened
- Use feedback to improve your final submission

Final Presentation Checklist

Before submitting your capstone, confirm that you have:

- ☐ Clearly explained the problem your system solves
- ☐ Presented your system architecture, behavior, and workflow
- ☐ Demonstrated how the system was evaluated using supporting evidence
- ☐ Clearly communicated the rationale behind your design
- ☐ Supported major design decisions with evidence
- ☐ Discussed engineering trade-offs
- ☐ Identified limitations and future improvements
- ☐ Prepared a clear, well-organized presentation

☐ Included a working version of your retrieval-augmented LLM system.

Key Takeaways

- Building an effective retrieval-augmented system is only part of the engineering process. Equally important is being able to clearly and confidently communicate the engineering story behind your system.
- Engineering decisions should be justified using evaluation results and supporting evidence rather than intuition.
- Every AI system involves trade-offs that must be balanced according to the intended use case.
- Feedback and iteration strengthen both systems and the way they are communicated.
- The ability to clearly explain and defend technical decisions is an essential professional skill that extends beyond this program.

